16 ACIDS, ALKALIS AND TITRATIONS

This chapter explores what indicators, acids and alkalis are, and how to carry out an acid–alkali titration. The reactions of acids are also discussed in Chapter 17.



▲ Figure 16.1 Acids range from the extremely dangerous, needing protective clothing to clean up spills . . .



▲ Figure 16.2 . . . to a natural part of our diet: oranges contain citric acid.

LEARNING OBJECTIVES

- Describe the use of litmus, phenolphthalein and methyl orange to distinguish between acidic and alkaline solutions
- Understand how the pH scale, from 0–14, can be used to classify solutions as strongly acidic (0–3), weakly acidic (4–6), neutral (7), weakly alkaline (8–10) and strongly alkaline (11–14)
- Describe the use of universal indicator to measure the approximate pH value of an aqueous solution
- Know that acids in aqueous solution are a source of hydrogen ions, and alkalis in an aqueous solution are a source of hydroxide ions
- Know that alkalis can neutralise acids

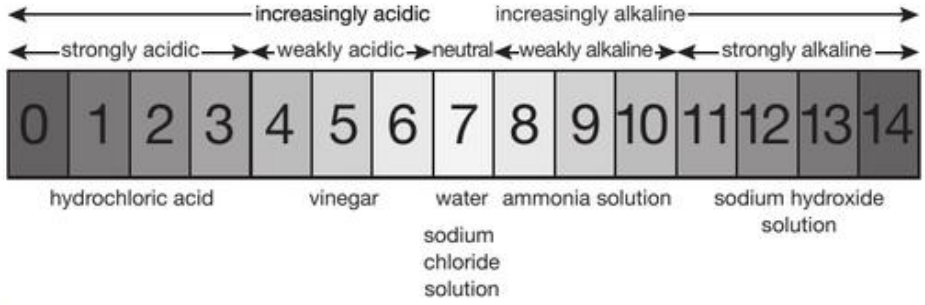
CHEMISTRY ONLY

- Describe how to carry out an acid–alkali titration.

pH AND INDICATORS

THE pH SCALE

The pH scale ranges from about 0 to about 14, and tells you how acidic or how alkaline a solution is.



▲ Figure 16.3 The pH scale. Vinegar is a solution of ethanoic acid.

We can classify substances as strongly acidic/alkaline or weakly acidic/alkaline based on their pH. Remember, when writing pH, it is a small p and a big H (the symbol for hydrogen).

Table 16.1 We can classify solutions according to their pH

	pH	Solution
strongly acidic	0–3	hydrochloric acid
weakly acidic	4–6	ethanoic acid (vinegar)
neutral	7	sodium chloride
weakly alkaline	8–10	ammonia
strongly alkaline	11–14	sodium hydroxide

EXTENSION WORK

These are actually only approximate ranges and depend on the concentration of the acid/alkali solution. A 0.10 mol/dm^3 solution of ethanoic acid actually has a pH of 2.88.

MEASURING pH

USING UNIVERSAL INDICATOR

HINT

You can measure pH much more accurately using a pH meter.

Universal indicator is made from a mixture of indicators, which change colour in a gradual way over a range of pH values. It can be used as a solution or as paper. The most common form is known as *full-range* universal indicator. It changes through a variety of colours from pH 1 right up to pH 14, but it isn't very accurate.



▲ Figure 16.4 Using universal indicator solution to measure the pH of various solutions.

The colour of the paper or solution is checked against a chart to find the pH.

ACID–ALKALI INDICATORS

Any substance that has different colours depending on the pH can be used as an indicator. Some common indicators and their colours in acidic and alkaline solutions are shown in Table 16.2.

Table 16.2 The colours of various indicators in acidic and alkaline solutions

	Acid	Alkali
litmus	red	blue
methyl orange	red	yellow
phenolphthalein	colourless	pink
universal indicator	red	blue

KEY POINT

Litmus can be used as a solution or as litmus paper.

Litmus is red in acidic solutions and blue in alkaline ones. In neutral solutions, the colour is purple, which is an equal mixture of the red and blue forms. Universal indicator is green in neutral solutions.

ACIDS

Table 16.3 Some acids showing the replaceable hydrogen

Acid	Formula
hydrochloric acid	HCl
nitric acid	HNO_3
sulfuric acid	H_2SO_4
ethanoic acid	CH_3COOH
phosphoric acid	H_3PO_4

The formulae of some acids are given in Table 16.3.

All acids contain hydrogen and when acids react the hydrogen shown in red in Table 16.3 is replaced by something else; all acids have replaceable H. For example, when hydrochloric acid reacts with sodium hydroxide we obtain:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

The H of the HCl has been replaced by an Na.

Not all of the hydrogens in acids are replaceable, for example in ethanoic acid only the H attached to the O is replaceable, not the ones joined to the C. The reactions of carboxylic acids such as ethanoic acid are discussed further in Chapter 27.

When acids are in water they dissociate (break apart) to form hydrogen ions (H^+), for example:

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$
$$HNO_3(aq) \rightarrow H^+(aq) + NO_3^-(aq)$$
$$H_2SO_4(aq) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$$

EXTENSION WORK

At the next level, you will discover that pH is defined by a mathematical equation: $pH = -\log_{10}[H^+(aq)]$.

When we are measuring pH we are actually measuring the concentration of these H^+ ions in the solution. This is why the H in pH is written with a capital letter.

We can define acids as *substances that act as a source of hydrogen ions (H^+) in solution*.

BASES

Bases are substances that neutralise acids by combining with the hydrogen ions in them. When we are referring to a base at this level we usually mean a metal oxide, a metal hydroxide or ammonia (NH_3).

ALKALIS

Some bases dissolve in water to form solutions containing hydroxide ions. These are alkalis.

Alkalis are a source of hydroxide (OH^-) ions in solution.

Examples of alkalis are sodium hydroxide and potassium hydroxide (all the Group 1 hydroxides). When sodium hydroxide is in water it breaks apart to form sodium and hydroxide ions:

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

The other alkali you will meet is a solution of ammonia (NH_3). The ammonia reacts with the water to form ammonium ions and hydroxide ions:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

These alkalis all have a pH greater than 7.



▲ Figure 16.5 Ammonia solution is sometimes called ammonium hydroxide. It is an alkali and is used in some cleaning products. Some universal indicator solution has been added to it in the beaker.

OTHER ALKALINE SOLUTIONS

There are some other substances, such as soluble metal carbonates, that react with water to form hydroxide ions:

$$Na_2CO_3(aq) + H_2O(l) \rightleftharpoons NaOH(aq) + NaHCO_3(aq)$$

EXTENSION WORK

The ionic equation for a carbonate reacting with water is:

$CO_3^{2-}(aq) + H_2O(l) \rightleftharpoons OH^-(aq) + HCO_3^-(aq)$

There are not many soluble carbonates, but sodium carbonate and potassium carbonate are both alkalis with a pH greater than 7. This is due to the OH^- ions in the solution. Only some of the carbonate ions react with water, so these solutions are only weakly alkaline.

REACTING ACIDS WITH BASES AND ALKALIS

Acids react with bases or alkalis in a neutralisation reaction.

REACTING ACIDS WITH BASES

Metal oxides, such as copper(II) oxide and magnesium oxide, are bases.



▲ Figure 16.6 Copper(II) oxide reacts with hot dilute sulfuric acid to form blue copper(II) sulfate solution.

Copper(II) oxide reacts with hot dilute sulfuric acid, in a neutralisation reaction, to produce a solution of copper(II) sulfate and water:

$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

Copper(II) oxide is an ionic compound containing the O^{2-} ion and what has happened in this reaction is that the H^+ ions from the acid have combined with the O^{2-} ions from the base to form water (H_2O).

REACTING ACIDS WITH ALKALIS

Sodium hydroxide solution (an alkali) reacts with dilute hydrochloric acid to form sodium chloride and water:

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

This is a neutralisation reaction. Sodium hydroxide and sodium chloride are both ionic compounds and so will be present as ions in solution. HCl(aq) is an acid and so will dissociate into $H^+(aq)$ and $Cl^-(aq)$. We can therefore re-write this equation showing all the ions.

$$Na^+(aq) + OH^-(aq) + H^+(aq) + Cl^-(aq) \rightarrow Na^+(aq) + Cl^-(aq) + H_2O(l)$$

The $Na^+(aq)$ and $Cl^-(aq)$ ions are the same on both sides of the equation and so are spectator ions. We can leave the spectator ions out of this equation to just show the things that have changed:

$$OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$$

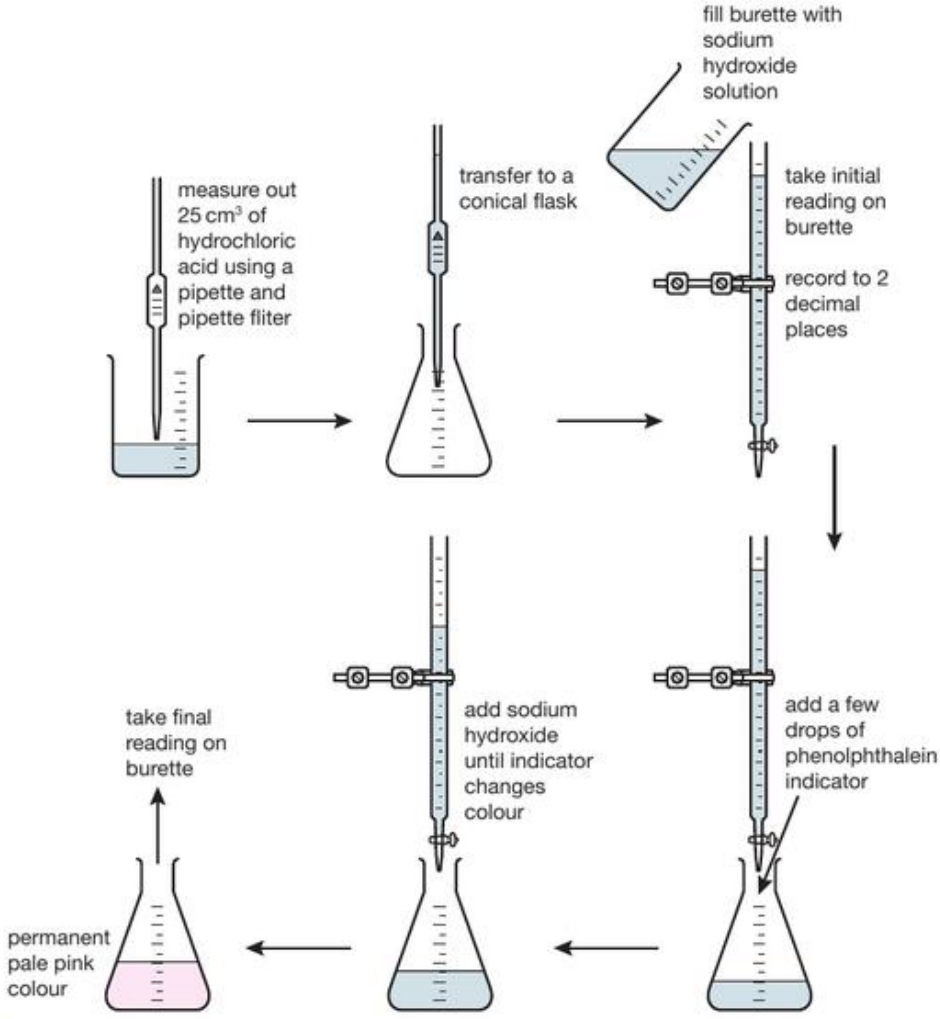
This is an ionic equation. All neutralisation reactions for an acid reacting with an alkali will have the same ionic equation. This is because they all involve the OH^- ions from the alkali reacting with the H^+ ions from the acid to form water.

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TITRATION

We can use titration to follow the course of a neutralisation reaction between an acid and alkali. Titration can be used to find out how much of the acid/alkali reacts with a certain volume of the alkali/acid.

The technique for carrying out a titration using phenolphthalein as indicator is shown in Figure 16.7.

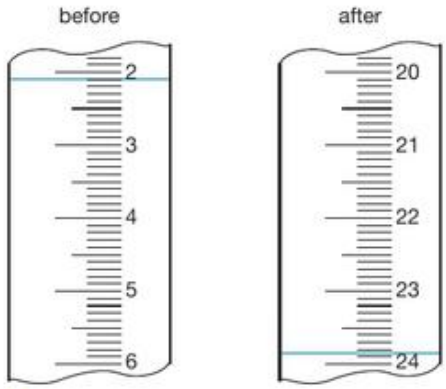


▲ Figure 16.7 How to carry out an acid–alkali titration

Calculations involving titrations are discussed in Chapter 6.

The alkali is added to the acid from the burette until the indicator changes colour. We usually do a rough titration first in order to find out approximately how much alkali is required to neutralise the acid. Imagine we found out that we required 22.50 cm^3 of alkali in the rough titration. We would then do the experiment again but this time running in about 20 cm^3 of the alkali quite quickly, swirling the conical flask all the time to mix the solutions. We would then add the alkali very slowly (dropwise) until one drop of alkali causes the indicator to change from its acid to its alkali colour. We then know the amount needed for neutralisation to within 1 drop; this is about as precise as we can be. In the above titration we could also have added the acid to the alkali but it is easier to see this indicator turning pink rather than going colourless.

When reading a burette it is important to remember that the numbers increase from the top to the bottom: 0 is at the top and 50 is at the bottom. We normally record readings from the burette to the nearest 0.05 cm^3, therefore all readings should be written down to 2 decimal places. The second decimal place is given as '0' if the level of solution is on the line and '5' if it is between the lines. The readings from the burette in Figure 16.8 should be recorded as shown in Table 16.4.



▲ Figure 16.8 Reading a burette

KEY POINT

The endpoint of a titration is the point at which the indicator changes colour.

The point at which the indicator changes colour does not necessarily indicate the point at which the solution is neutral. The pH changes very rapidly near the endpoint of the titration and in some titrations 1 drop of an alkali can change the pH from about 3 to 11. So the most precise we can be is to know the neutralisation point to within 1 drop.

Table 16.4 Results table from a titration experiment

final reading on burette/cm^3	23.85
initial reading on burette/cm^3	2.10
volume of alkali added/cm^3	21.75

The volume of alkali added is obtained by subtracting the initial reading from the final reading:

volume of alkali = 23.85 – 2.10 = 21.75 cm^3

Various other indicators can also be used, such as methyl orange, but not universal indicator. We do not use universal indicator because it has a range of colours and changes gradually between them; this means we would not be able to see a clear endpoint for the titration.

END OF CHEMISTRY ONLY

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 4

1 State the colours of methyl orange, phenolphthalein and litmus in each of the following solutions:
a sodium hydroxide solution b hydrochloric acid solution.

SKILLS ANALYSIS

2 Table 16.5 gives the pH of some solutions. Classify each as strongly acidic, strongly alkaline, weakly acidic, weakly alkaline or neutral by copying the table and putting ticks in the appropriate boxes.

Table 16.5

Solution	pH	Strongly acidic	Weakly acidic	Neutral	Weakly alkaline	Strongly alkaline
potassium iodide	7					
propanoic acid	4.2					
sodium carbonate	9.5					
potassium hydroxide	13					
iron(III) chloride	2.4					
nitric acid	1.3					

SKILLS PROBLEM SOLVING 9

3 Write equations for the reactions between:
a sodium oxide and nitric acid b potassium hydroxide and sulfuric acid.

SKILLS CRITICAL THINKING 6

4 Explain what type of reaction is occurring in 3.

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5 A student carried out an acid–alkali titration. They used the following set of instructions. Copy out the paragraph and fill in the blanks.

Measure out 25.0 cm^3 of potassium hydroxide solution using a _____. Transfer the potassium hydroxide solution to a _____ _____. Add a few drops of an _____. Put sulfuric acid into the _____. Add the acid to the alkali until the _____ _____ _____.

END OF CHEMISTRY ONLY